E2: CMX + Cross-Modal Residual Filtering (CMRF)

What is different from E1, and why we chose to improve this one factor

Team 41 · RGB-thermal semantic segmentation on PST900 · Baseline: CMX (MiT-B2)

The crux in one paragraph

E1 showed that CMX, although built to fuse RGB and thermal, behaves almost like an RGB-only model on PST900. Damaging the thermal image costs at most 1.3 mIoU, while damaging RGB costs up to 66.6 mIoU. Under severe RGB darkness the model detects almost nothing (object mIoU 0.17%), even though the thermal image is intact. **So the one factor E2 targets is the unused thermal information.** CMRF predicts each camera's features from the other, keeps only the part that *cannot* be predicted (the information unique to that camera), and gates it into the fusion. Everything else in E1 stays identical, so any change in the results can be attributed to this single module.

Why one factor only

- **Scientific control.** E2 changes exactly one thing (CMRF plus its small auxiliary loss). Data, split, epochs, optimiser, learning rate, batch size, crop and evaluation are identical to E1. If results change, we know why.
- **Clean accuracy is not the real gap.** E1 reaches 86.12% mIoU. The best published PST900 result we reviewed, CRM (Swin-B, a larger model), reaches 88.0%: a gap of about 2 points on a saturated benchmark.
- **Robustness is the real gap.** Under RGB degradation E1 falls to 67.4% on average and to 19.6% at the strongest low-light level: a gap of up to 66 points.
- **It is the supervisor's research question:** "*Can cross-modal feature prediction and selective residual filtering improve RGB-thermal segmentation robustness under modality-specific degradation while preserving clean-input accuracy and limiting computational overhead?*"

1. How we arrived at this direction

The decision followed a chain of evidence rather than a guess. Each step below is backed by a measured result or by a paper we read.

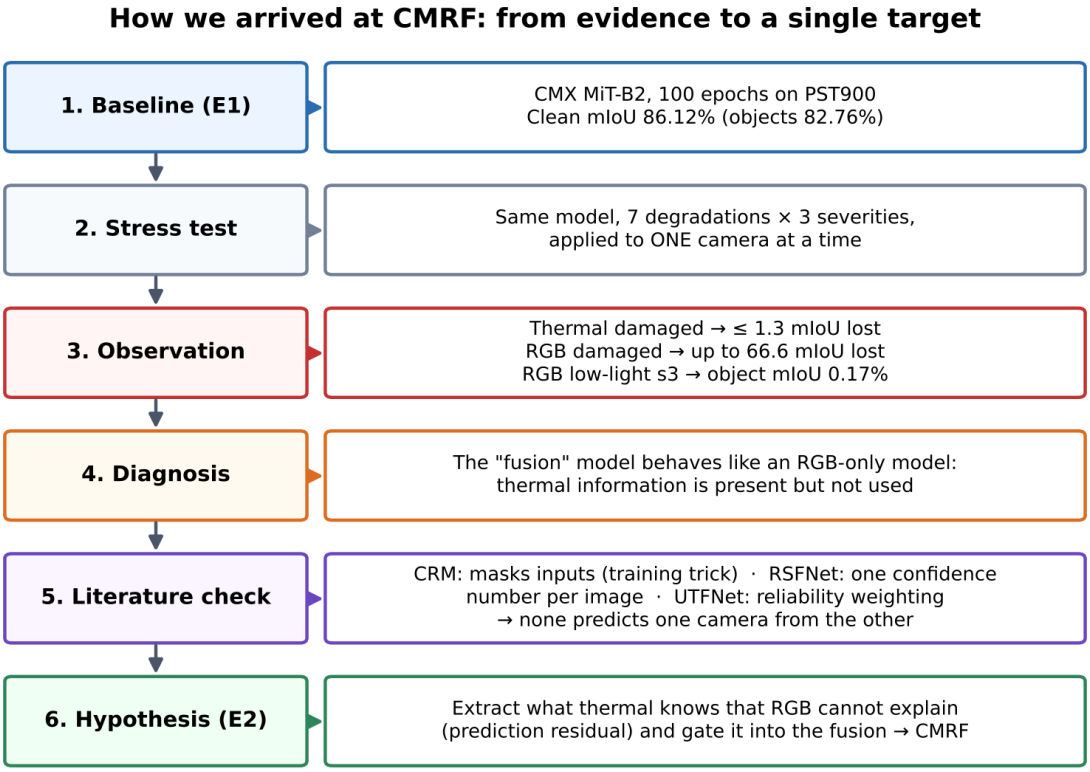


Figure 1. From the E1 baseline to the E2 hypothesis.

2. The evidence from E1

E1 is the official CMX (MiT-B2) trained on PST900 (597 training / 288 test pairs, 5 classes) for 100 epochs on a T4 GPU, batch size 4, AdamW with learning rate 6e-5. It was evaluated with sliding-window inference on the original 720×1280 labels, first on clean images and then with 7 degradations × 3 severities, each applied to **one camera at a time**.

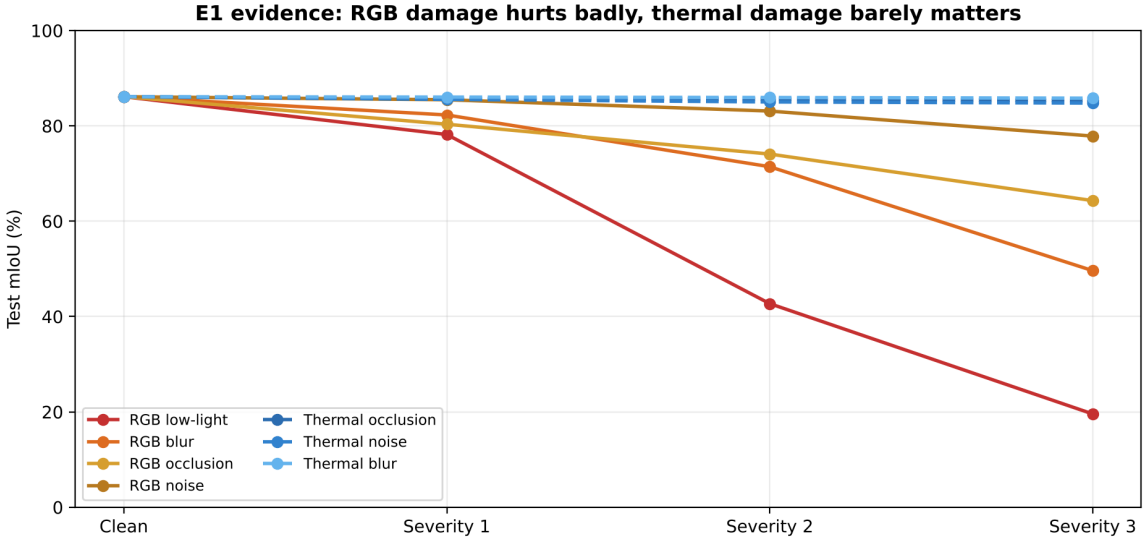


Figure 2. E1 test mIoU as each degradation gets stronger (solid = RGB damaged, dashed = thermal damaged).

Measurement (E1)	Value	What it tells us
Clean mIoU (5 classes / 4 objects)	86.12% / 82.76%	A solid, believable baseline
Weakest class	Survivor: IoU 78.8, sensitivity 82.2%	About 18% of survivor pixels are missed
Thermal occlusion, severity 3	85.32 (−0.8)	About 35% of the thermal image covered: barely noticed
RGB occlusion, severity 3	64.32 (−21.8)	Same damage on RGB: about 27× larger effect
Average under RGB / thermal damage	67.42 / 85.53	The model depends on RGB
RGB low-light, severity 3 (thermal intact)	mIoU 19.57; object mIoU 0.17%	Thermal does not rescue the model
Cost	66.56 M params, 78.27 GFLOPs, ≈9.1 FPS (T4)	Reference for E2's overhead

The qualitative figure from E1 shows the same pattern: under RGB low-light the predictions are completely empty, while under thermal occlusion they are almost identical to the clean predictions.

3. Why CMRF, and not an existing method

We checked whether existing work already solves this. Three methods address related ideas, but none extracts the information *unique* to each camera by predicting one camera from the other.

Method	Where it acts	Key idea	Relation to our problem
CMX (baseline)	Features, at every stage	Rectify each modality using the other (CM-FRM), then fuse (FFM)	No mechanism to switch to thermal when RGB fails: this is what E1 exposed
CRM (Shin et al., ICRA 2024)	Input pixels, training only	Complementary patch masking, modality-wise supervision, self-distillation (clean vs masked); max fusion	Same goal (avoid over-reliance), different tool. Robustness tested only as full modality drop, on MFNet
RSFNet (Li et al., 2023)	Feature fusion	"Residual" = skip connection; gate = **one confidence number per image** learned from saliency pseudo-labels	Not a prediction residual; no robustness experiments
UTFNet (GRSL 2023)	Fusion	Uncertainty-guided ("trustworthy") fusion	Full text not yet read (IEEE access pending); to be verified
CMRF (ours, E2)	**Features, before CMX's CM-FRM**	**Predict one camera from the other; keep the unexplained residual; gate it per pixel and channel**	**Directly targets the unused thermal information**

The difference in one line. CRM changes *how the model is trained*; RSFNet and UTFNet decide *how much* to trust each camera; CMRF extracts *what new information* each camera brings, using the cross-modal prediction error as a free, label-free signal.

4. What is different from E1

Item	E1	E2
Model	CMX (MiT-B2)	CMX (MiT-B2) **+ CMRF at encoder stage 3**
Training loss	Cross-entropy	Cross-entropy **+ 0.1 × cross-modal prediction loss**
Code changes	–	`models/encoders/dual_segformer.py` (CMRF before `FRMs[2]`), `models/builder.py` (adds the auxiliary loss), new `models/cmrf.py`
Extra parameters / GFLOPs	–	**+0.465 M (+0.70%)** / **+0.56 (+0.71%)**, measured at 480×640
Dataset, split, epochs (100), optimiser, LR, batch (4), crop (480×640)	Same	**Same**
Evaluation (clean, 7 degradations × 3 severities, cost)	Same	**Same**, plus E1 vs E2 tables, modality-drop test and gate maps

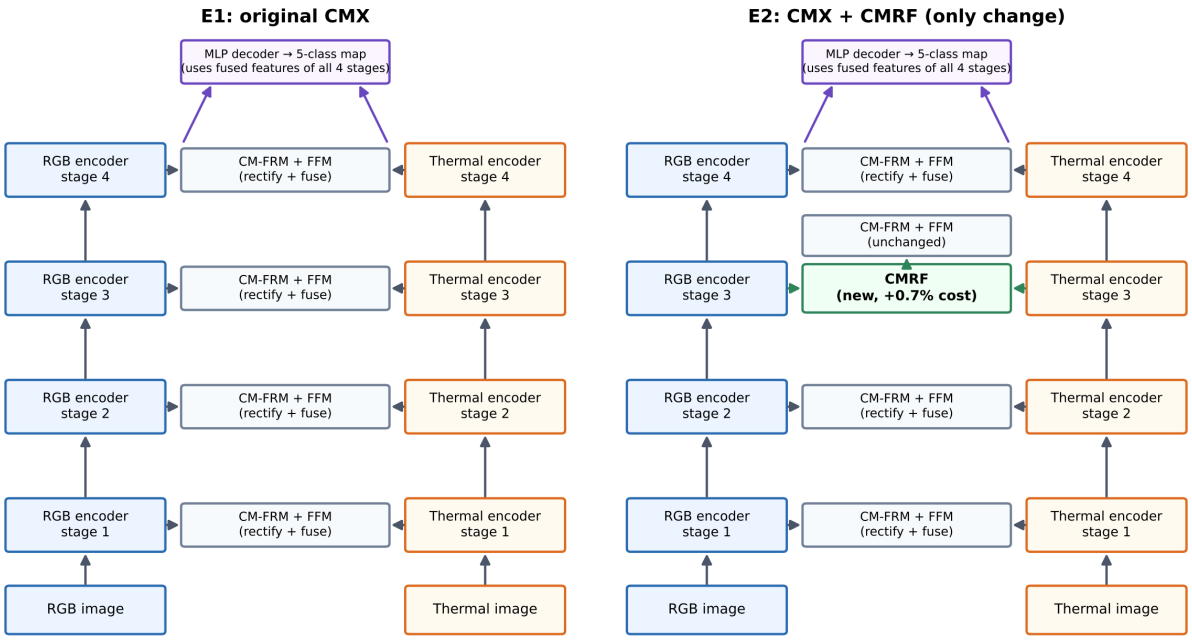


Figure 3. E1 (left) and E2 (right). CMRF is inserted only at stage 3, before CMX's own rectification and fusion, which stay unchanged.

5. How CMRF works

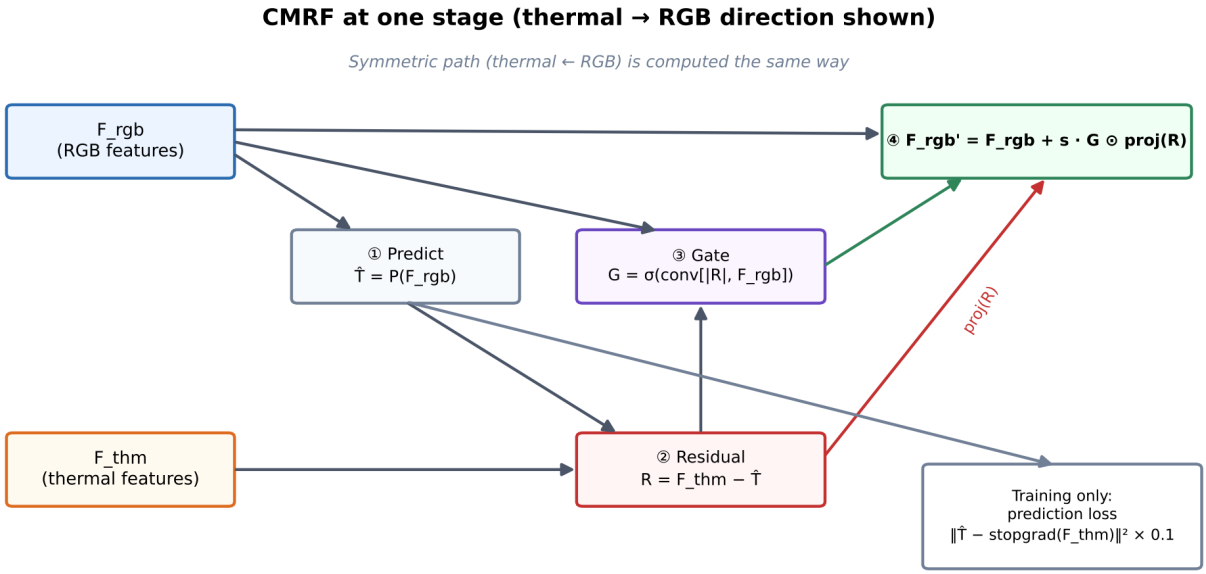


Figure 4. CMRF at one stage, thermal-to-RGB direction. The RGB-to-thermal direction is computed in the same way.

For the RGB features F_{rgb} and thermal features F_{thm} at stage 3:

Thermal → RGB path

$\tilde{T} = P_{r\to t}(F_{rgb})$

$Res_t = F_{thm} - \tilde{T}$

$G_t = \sigma(\ conv[\ |Res_t|\ ,\ F_{rgb}\ ]\ )$

$F_{rgb}' = F_{rgb} + s_t \cdot G_t \odot proj(Res_t)$

RGB → thermal path (same form)

$\tilde{R} = P_{t\to r}(F_{thm})$, $Res_r = F_{rgb} - \tilde{R}$, $F_{thm}' = F_{thm} + s_r \cdot G_r \odot proj(Res_r)$

Training loss

$L = L_{CE} + 0.1 \cdot [\ NMSE(\tilde{T}, sg(F_{thm})) + NMSE(\tilde{R}, sg(F_{rgb}))\ ]$

Here sg(·) is stop-gradient, NMSE is mean squared error divided by the target's mean square, and ⊙ is element-wise multiplication.

Why this should help the E1 failure

- When RGB is dark, the RGB features carry little information, so thermal features **cannot be predicted** from them. The thermal residual becomes large exactly where RGB failed, giving the gate a natural signal to pass thermal information into the RGB stream.
- When both cameras agree, the residual is small and CMRF changes little, so clean accuracy should be preserved.
- The signal comes from the prediction error itself. It needs no extra labels (unlike RSFNet's saliency pseudo-labels) and no explicit uncertainty model.

6. Design choices and why

Choice	Value	Reason
Stage	3 only	Supervisor's plan: start at a single stage. Stage 3 features are semantic (objects) but still spatial (30×40 at 480×640 input). Stage 2 is the planned ablation
Predictor	1×1 → depthwise 3×3 → 1×1, width = channels/4	Lightweight (total +0.70% params); 3×3 adds local context
Gate	Per pixel and per channel, sigmoid	Localised degradations need local decisions; RSFNet uses one number per image
Gate input	[|residual|, receiving features]	The size of the unexplained part plus the context it would be added to
Prediction loss	NMSE with stop-gradient target, λ = 0.1	Makes predictors learn real correspondence; stop-gradient prevents feature collapse
Injection strength s	Learned, starts at 0.1	Starts close to E1 (stable fine-tuning of pretrained weights) but can grow
Direction	Both (thermal→RGB and RGB→thermal)	Symmetric; the data decides which direction matters

7. How to judge E2

Metric	E1	What would support the hypothesis
Clean mIoU	86.12	≈ 86 (no meaningful drop)
Average mIoU under RGB degradation	67.42	Higher than E1
RGB low-light, severity 3	19.57 (objects 0.17)	Clear recovery
Average under thermal degradation	85.53	Slightly lower is acceptable: a sign the model now uses thermal
Modality drop (thermal only)	to be measured	Higher than E1: thermal branch more useful
Extra cost	–	About +0.7% params and GFLOPs (measured)
Gate maps	–	Stronger gate on objects when RGB is dark

What E2 can and cannot show

- **E2 is trained only on clean images.** The model never sees dark RGB during training, so its robustness gain may be modest. That is expected: **E3** adds modality-specific corruption during training, and that is where a larger gain is expected.
- **The control run is essential.** E1 + the same corruption training, *without* CMRF, separates CMRF's contribution from the effect of the augmentation.
- **CMRF is still a hypothesis.** As the supervisor asked, its effect must be shown through ablations (stage, gate on/off, residual vs plain prediction, prediction loss on/off), and its novelty verified (UTFNet full text, cross-modal reconstruction and missing-modality literature).
- **Known risks:** the gate may stay near-constant (check the gate maps and the learned scale); the predictor may be too weak to make the residual meaningful; E4's consistency loss is close to CRM's self-distillation and should be presented as adapted from it.

8. Running E2 and what it produces

- Open `AIProj_CMX_PST900_E2_CMRF.ipynb` in Colab with a GPU runtime and choose Run all. It uses the same Drive folders, dataset zip and pretrained weights as E1; the run name is `E2_cmrf_s3_ep100`, so E1 files are not touched.
- New sections: **8b** adds CMRF; **11b** checks the model and its overhead; **15b** builds E1 vs E2 tables and plots; **16b** runs the modality-drop test; **16c** shows the residual and gate maps under clean and RGB low-light.
- Training takes about the same time as E1 (about 4.8 h for 100 epochs on a T4). Disconnects are handled by resuming from the latest checkpoint.
- Output: `E2_results.zip` with all CSVs (including `E1_vs_E2_*.csv`), figures, CMRF maps and the training log; the trained model is saved as `E2_cmrf_s3_ep100_final.pth`.

9. Next steps after E2

- **E3:** E2 + corruption of one modality at a time during training, plus the control (E1 + the same corruption).
- **E4:** E3 + clean-to-corrupted consistency (adapted from CRM's self-distillation).
- **Ablations:** CMRF at stage 2 / stages 2+3; gate removed; prediction loss removed.
- **Repeat key runs with 2–3 seeds** before claiming small differences.